§ 11 Perturbation

1. RS

$H|n\rangle = E_n|n\rangle$

$$\begin{cases} H = H_0 + V \\ E_n = E_n^0 + E_n^1 + \cdots \\ |n\rangle = |n^0\rangle + |n^1\rangle + \cdots \end{cases} \qquad (H_0+V)(|n^0\rangle + |n^1\rangle + \cdots) = (E_n^0 + E_n^1 + \cdots)(|n^0\rangle + |n^1\rangle + \cdots)$$

0階: $H_0|n^0\rangle = E_n^0|n^0\rangle$

1階: $H_0|n^1\rangle + V|n^0\rangle = E_n^0|n^1\rangle + E_n^1|n^0\rangle$

$\langle n^0|$: $E_n^0\langle n^0|n^1\rangle + \langle n^0|V|n^0\rangle = E_n^0\langle n^0|n^1\rangle + E_n^1 \Rightarrow E_n^1 = \langle n^0|V|n^0\rangle$

$\langle m^0|$: $E_m^0\langle m^0|n^1\rangle + \langle m^0|V|n^0\rangle = E_n^0\langle m^0|n^1\rangle + E_n^1\langle m^0|n^0\rangle$

歸一化條件: $(\langle n^0| + \langle n^1|)(|n^0\rangle + |n^1\rangle) = \langle n^0|n^0\rangle + \langle n^1|n^0\rangle + \langle n^0|n^1\rangle + \mathcal{O}(2)$

$= 1 + \langle n^1|n^0\rangle + \langle n^1|n^0\rangle^* = 1$

if $\langle n^1|n^0\rangle \in \mathbb{R} \Rightarrow \langle n^1|n^0\rangle = 0$

$$|n^1\rangle = \left(\sum_m |m^0\rangle\langle m^0|\right)|n^1\rangle = |n^0\rangle\langle n^0|n^1\rangle + \sum_{m\neq n} |m^0\rangle\langle m^0|n^1\rangle$$

$$= \sum_{m\neq n} |m^0\rangle\langle m^0|n^1\rangle$$

2階: $H_0|n^2\rangle + V|n^1\rangle = E_n^0|n^2\rangle + E_n^1|n^1\rangle + E_n^2|n^0\rangle$

$\langle n^0|$:

$\langle m^0|$:

2. Example : Two-level System

$H = B_z \sigma_z + \varepsilon \sigma_x = H_0 + V$

$E_n^0 : \begin{cases} B_z : |1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ -B_z : |2\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{cases}$

$E_1^1 = \langle 1|V|1\rangle = 0$, $E_2^1 = \langle 2|V|2\rangle = 0$

$|1'\rangle =$

Example 2 :

$H = \frac{1}{2} m \omega^2 x^2 \rightarrow \qquad \delta x^4$

3. BW.

$(E_n - H_0)|n\rangle = V|n\rangle$

$\langle m^0|E_n - H_0|n\rangle = \langle m^0|V|n\rangle \quad \rightarrow \langle m^0|n\rangle = \frac{\langle m^0|V|n\rangle}{E_n - E_m^0}$

$|n\rangle = \left(|n^0\rangle\langle n^0| + \sum_{m \neq n} |m^0\rangle\langle m^0|\right)|n\rangle$

$= |n^0\rangle\langle n^0|n\rangle + \sum_{m \neq n} \frac{\langle m^0|V|n\rangle}{E_n - E_m^0}|m^0\rangle$

$\sum = 1$

$= |n^0\rangle + \sum_{m \neq n} \frac{\langle m^0|V|n\rangle}{E_n - E_m^0}|m^0\rangle$

$\langle n^0|E_n - H_0|n\rangle = \langle n^0|V|n\rangle \quad \rightarrow E_n = E_n^0 + \frac{\langle n^0|V|n\rangle}{\langle n^0|n\rangle} = E_n^0 + \langle n^0|V|n\rangle$

$E_n = E_n^0 + \langle n^0|V|n\rangle$

$\approx E_n^0 + \langle n^0|V|n^0\rangle + \sum_{m \neq n} \frac{\langle n^0|V|m^0\rangle\langle m^0|V|n^0\rangle}{E_n - E_m^0}$

$|n\rangle \approx |n^0\rangle$

Application to Continuum : Resonance State.

$$E_n^0 + \langle n^0|V|n^0\rangle + \sum_{m\neq n} \frac{\langle n^0|V|m^0\rangle\langle m^0|V|n^0\rangle}{E_n - E_m^0} \qquad |m\rangle \approx |n^0\rangle$$

$$\downarrow$$

$$E_n = E_n^0 + \langle n^0|V|n^0\rangle + \int d\varepsilon\, \rho(\varepsilon) \frac{|\langle \varepsilon|V|n\rangle|^2}{E_n - \varepsilon + i\eta} \qquad \frac{1}{x+i\eta} = \frac{P}{x} - i\pi\delta(x)$$

$$= E_n^0 + \langle n^0|V|n^0\rangle + \int d\varepsilon\, \rho(\varepsilon)\, |V_{\varepsilon n}|^2 \left(\frac{P}{E_n - \varepsilon} - i\pi\, \delta(E_n - \varepsilon) \right)$$

$$= E_n^0 + \langle n^0|V|n^0\rangle + \int d\varepsilon\, \rho(\varepsilon)\, |V_{\varepsilon n}|^2 \frac{P}{E_n - \varepsilon} - i\pi\, \rho(E_n)\, |V_{nn}|^2$$

为演化中，$e^{-i(E_n - \frac{i}{2}\Gamma)t} = e^{-\frac{\Gamma}{2}t} e^{-iE_n t}$

对应衰减

$$|n\rangle = |n^0\rangle + \int d\varepsilon\, \rho(\varepsilon) \frac{|V_{\varepsilon n}|}{\mathrm{Re}[E_n] - \varepsilon - i\frac{\Gamma}{2}}$$

4. Fine Structure of H Atom. Zeeman Effect.

$$E_{nj} = -\frac{1}{2}mc^2\left(\frac{\alpha}{n}\right)^2 + \frac{1}{2}mc^2\left(\frac{\alpha}{n}\right)^4\left(\frac{3}{4} - \frac{n}{j+\frac{1}{2}}\right)$$

$$H = H_H + H_Z, \quad H_Z = \mu_B(L_z + 2S_z)B_z$$

$|l, m_l\rangle \Leftrightarrow |l, m_l, s, m_s\rangle$

① Strong B.

$H_0 = \mu_B(L_z + 2S_z)B_z$

$E_Z = \langle l, m_l, s, m_s | H_0 | l, m_l, s, m_s\rangle = \mu_B B_z(m_l + 2m_s)\hbar$

选取 $|l, m_l, s, m_s\rangle$ 基矢

$|l=1, m_l=1, \uparrow\rangle$

$|l=1, m_l=0, \uparrow\rangle$

$|l=1, m_l=0, \downarrow\rangle$

$|l=1, m_l=-1, \downarrow\rangle$

$|j=\frac{3}{2}, m_j=\frac{3}{2}, l=1\rangle$

$|j=\frac{3}{2}, m_j=\frac{1}{2}, l=1\rangle$

$|j=\frac{3}{2}, m_j=-\frac{1}{2}, l=1\rangle$

$|j=\frac{3}{2}, m_j=-\frac{3}{2}, l=1\rangle$

2

1

1

② Weak B

$V = \mu_B(L_z + 2S_z)B_z$

$E_Z = \langle j, m_j, l | V | j, m_j, l\rangle$

$= g_{jl}\mu_B B_z \langle j, m_j, l | J_z | j, m_j, l\rangle = m_{jl}\mu_B B_z \hbar$

$$g_{jl} = 2 - \frac{j(j+1) + l(l+1) - \frac{3}{4}}{2j(j+1)}$$

Stark Effect

6. High Order Degenerate Perturbation

$$H_0 = \begin{pmatrix} 0 & 0 \\ 0 & \Delta \end{pmatrix}, \quad V = \begin{pmatrix} V_1 & \\ V_2 & \\ V_1 V_2 & 0 \end{pmatrix}$$

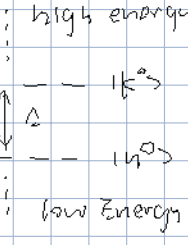
$E_1^0 = E_2^0 \to \infty$

$$E_n^{(2)} = \sum_{m} \frac{\langle m^0|V|n^0\rangle\langle m^0|V|n^0\rangle}{E_n^0 - E_m^0} \Rightarrow \begin{cases} E_1^{(2)} = -\frac{V_1^2}{\Delta} \\ E_2^{(2)} = -\frac{V_2^2}{\Delta} \end{cases}$$

$\Rightarrow$ Eff Hamiltonian

$$\hat{P} := \sum_n |n^0\rangle\langle n^0| \qquad P+Q=1$$

$$\hat{Q} := \sum_k |k^0\rangle\langle k^0| \qquad P^2 = P, \; Q^2 = Q$$



$$\hat{H}|\psi\rangle = E|\psi\rangle$$

$$\hat{P}\hat{H}|\psi\rangle = E\hat{P}|\psi\rangle, \qquad \hat{Q}\hat{H}|\psi\rangle = E\hat{Q}|\psi\rangle$$

$$\hat{P}\hat{H}(\hat{P}+\hat{Q})|\psi\rangle = E\hat{P}|\psi\rangle \qquad \hat{Q}\hat{H}(\hat{Q}+\hat{P})|\psi\rangle = E\hat{Q}|\psi\rangle$$

$$\hat{P}\hat{H}\hat{P}\hat{P}|\psi\rangle + \hat{P}\hat{H}\hat{Q}\hat{Q}|\psi\rangle = E\hat{P}|\psi\rangle, \qquad \hat{Q}\hat{H}\hat{Q}\hat{Q}|\psi\rangle + \hat{Q}\hat{H}\hat{P}\hat{P}|\psi\rangle = E\hat{Q}|\psi\rangle$$

$$\hat{P}\hat{H}\hat{P}|\psi\rangle + \hat{P}\hat{H}\hat{Q}\left[E - \hat{Q}\hat{H}\hat{Q}\right]^{-1}\hat{Q}\hat{H}\hat{P}|\psi\rangle = E\hat{P}|\psi\rangle$$

$$H_{eff} := \hat{P}\hat{H}\hat{P} + \hat{P}\hat{H}\hat{Q}\underbrace{\left[E - \hat{Q}\hat{H}\hat{Q}\right]^{-1}}_{\approx E_0}\hat{Q}\hat{H}\hat{P}$$

$$\hat{H}_{eff}|\psi_P\rangle = E_{eff}|\psi_P\rangle$$

$$H = \begin{pmatrix} H_H & V \\ V^\dagger & H_L \end{pmatrix} = \begin{pmatrix} QHQ & QHP \\ PHQ & PHP \end{pmatrix}$$

$$\Rightarrow H_{eff} = H_L + V^\dagger \frac{1}{E - H_H} V \approx H_L + V^\dagger \frac{1}{E_0 - H_H} V$$

$$H_{eff} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} + (V_1 \; V_2)\frac{1}{0-\Delta}\begin{pmatrix} V_1 \\ V_2 \end{pmatrix} = -\frac{1}{\Delta}\begin{pmatrix} V_1^2 & V_1V_2 \\ V_1V_2 & V_2^2 \end{pmatrix}$$

변분법과 $E_0 = E_{eff}$

$$\Rightarrow E_1 = 0, \quad E_2 = -\frac{1}{\Delta}(V_1^2 + V_2^2)$$

$$\text{고유벡터} \; |1\rangle = \frac{1}{\sqrt{V_1^2+V_2^2}}\begin{pmatrix} V_2 \\ -V_1 \end{pmatrix}, \quad |2\rangle = \frac{1}{\sqrt{V_1^2+V_2^2}}\begin{pmatrix} V_1 \\ V_2 \end{pmatrix}$$